

```

#include<iostream>
#include<stdlib.h>
#include<GL/glut.h>
using namespace std;
#define s 0.86602540
int n;
void koch(int x1,int y1,int x2,int y2,int m)
{
    int x[5],y[5];
    x[0] = x1;
    y[0] = y1;
    x[4] = x2;
    y[4] = y2;
    int lx = (x2-x1)/3;
    int ly = (y2-y1)/3;
    x[1] = x[0] + lx ;
    y[1] = y[0] + ly ;
    x[3] = (x[0] + (2*lx));
    y[3] = y[0] + (2*ly);
    int xx = x[3]-x[1];
    int yy = y[3]-y[1];
    x[2] = x[1]+(xx*(0.5) + yy*s) ;
    y[2] = y[1]+(yy*(0.5) - xx*s) ;
    if(m>0)
    {
        koch(x[0],y[0],x[1],y[1],m-1);
        koch(x[1],y[1],x[2],y[2],m-1);
        koch(x[2],y[2],x[3],y[3],m-1);
        koch(x[3],y[3],x[4],y[4],m-1);
    }
    else
    {
        glBegin(GL_LINES);
        glVertex2d(x[0], y[0]);
        glVertex2d(x[1] , y[1] );
        glEnd();
        glBegin(GL_LINES);
        glVertex2d(x[1], y[1]);
        glVertex2d(x[2] , y[2] );
        glEnd();
        glBegin(GL_LINES);

```

```

glVertex2d(x[2], y[2]);
glVertex2d(x[3] , y[3] );
glEnd();
glBegin(GL_LINES);
glVertex2d(x[3], y[3]);
glVertex2d(x[4] , y[4] );
glEnd();
glFlush();
}
}
void display(void)
{
glClear(GL_COLOR_BUFFER_BIT);
glColor3f(1.0,1.0,1.0);
int x1=0,x2=550,y1=200,y2=200;
koch(x1,y1,x2,y2,n);
glFlush();
}
void myinit()
{
glClearColor(0.0,0.0,0.0,1.0);
glColor3f(1.0,1.0,1.0);
gluOrtho2D(0.0,650.0,650.0,0.0);
}
int main(int argc,char **argv)
{
cout<<"\nEnter the level of curve generation : ";
cin>>n;
glutInit(&argc,argv);
glutInitDisplayMode(GLUT_SINGLE|GLUT_RGB);
glutInitWindowSize(650,650);
glutInitWindowPosition(0,0);
glutCreateWindow("Koch Curve");
glutDisplayFunc(display);
myinit();
glutMainLoop();
return 0;
}

```